To my request, Gowarikar said: ‘Remember, you never recruited anyone from my team (solid propulsion) when you set out for Vernon.’

I wouldn’t have been surprised to hear such a thing from Gowarikar, the unit head; but when it came from Gowarikar, the VSSC director, I was a bit taken aback. Now I was convinced that Gowarikar would remain the old fox. The VSSC director was still thinking about ‘his team’, which was but a unit under the centre. I knew it is going to be a long haul, but with my constant nagging of Dhawan, I managed to hold on to a core team that continued with the making of the Vikas engine till it became a project and a part of PSLV.

By now my team was doing two things: putting together a Vikas engine using the original parts that went into the making of the French Viking engine; and indigenisation of materials and processes so that we can make more engines once we successfully tested the first one. Parallel to the two activities went another significant exercise: to re-decipher, analyse and plug holes in the ‘Vernon Document’.

This was but a compilation of reports our team had been sending to India while working in SEP, Vernon. Right from 1975 when we landed in Vernon, this exercise had begun. Our scientists would return from work every evening to one of the apartments that served as our office for a daily discussion and review. We had charted out ten phases of the Vernon project, starting from identification of sub-systems, detailed design and materials to fabrication, assembly, testing and integration of the stage.

Starting from a system, say the gas generator, our scientists would get down to sub-systems like injectors, regulators, combustion chamber and feed lines. We would then go down to each of the three types of regulators which had sub-sub-systems, such as ‘O’ rings, cylinders and pistons. An ‘O’ ring is a single component, and the breaking-down process ends there. If the cylinder has many parts, we go further dissecting to the last part.

Our scientists meticulously noted down every such single component of the Viking-Vikas engine. As we proceeded hand-in-hand with the French scientists, whose attitude towards us changed from curiousity and indifference to camaraderie and confidence, the Vernon Document gained muscle with details of materials, their behaviour, design methodology and fabrication process.

Some parts of this exercise, though elaborate and time-consuming, were easy; but some like the manufacturing process left us at dead-ends. These